

CHAPTER 6

$$6-1. (a) G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{0.5}{s(s+0.5)}\right] = \frac{z-1}{z} \frac{(1-e^{-1})}{(z-1)(z-e^{-1})} = \frac{0.6321}{z-0.3679}$$

$$\frac{G(z)}{1+G(z)} = \frac{0.6321}{z-0.2642}$$

$$\frac{C(z)}{z} = \frac{0.6321}{(z-1)(z+0.2642)} = \frac{0.5}{z-1} + \frac{-0.5}{z+0.2642}$$

$$\therefore c(kT) = 0.5[1 - (-0.2642)^k]$$

$$(b) \frac{G_p(s)}{1+G_p(s)} = \frac{0.5}{s+1}$$

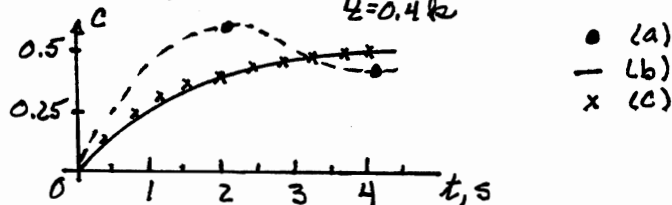
$$C(s) = \frac{0.5}{s(s+1)} = \frac{0.5}{s} - \frac{0.5}{s+1} \Rightarrow c(t) = 0.5(1 - e^{-t})$$

$$(c) G(z) = \frac{0.1813}{z-0.8187}, \quad \frac{G(z)}{1+G(z)} = \frac{0.1813}{z-0.6374}$$

$$\therefore \frac{C(z)}{z} = \frac{0.1813}{(z-1)(z-0.6374)} = \frac{0.5}{z-1} + \frac{0.5}{z-0.6374}$$

$$\therefore c(kT) = 0.5[1 - (0.6374)^k]$$

$$\text{In (b), } t = 0.4k, \text{ and } c(t)|_{t=0.4k} = 0.5[1 - e^{-0.4k}] = 0.5[1 - (0.6703)^k]$$



$$(d) (a) T(1) = 0.5 \quad (c) T(1) = 0.5$$

$$(b) T(0) = 0.5$$

dc gain is a function of only $G_p(0)$ and is independent of T .

$$6-2. (a) \tau = \frac{-T}{\ln r} = \frac{-2}{\ln(0.2642)} = 1.503 \text{ s}$$

$$(b) e^{-t/\tau} = 0.02 \quad \therefore t = -(1.503) \ln(0.02) = 5.88 \text{ s} \approx 4\tau$$

$$(c) (a) \tau = \frac{-0.4}{\ln(0.6374)} = 0.888 \text{ s}; (b) t = -\tau \ln(0.02) = 3.42 \text{ s} \approx 4\tau$$

$$(d) (a) \tau = 1 \text{ s}; (b) t = -\tau \ln(0.02) = 3.91 \text{ s} \approx 4\tau$$

6-3. (a) From Problem 6-1(a): $G(z) = \frac{0.6321}{z - 0.3678}$

From Example 6.1: $C(s) = G(s) \frac{R(z)}{1+G(z)} \Big|_{z=e^{Ts}}$

$$\frac{1}{1+G} = \frac{z - 0.3678}{z + 0.2642} = 1 - 0.632z^{-1} + \dots$$

$$C_1(s) = \frac{0.5}{s(s+0.5)} = \frac{1}{s} - \frac{1}{s+0.5}$$

$$C_1(t) = (1 - e^{-0.5t})$$

$$\therefore C(t) = (1 - e^{-0.5t})u(t) - 0.632(1 - e^{-0.5(t-2)})u(t-2) + \dots$$

$$\therefore C(t) \Big|_{t=1} = 1 - e^{-0.5} = \underline{0.3935}$$

$$(b) C(t) \Big|_{t=3} = 1 - e^{-1.5} - 0.632(1 - e^{-0.5}) = \underline{0.5281}$$

(c) The output $C(t)$ cannot change instantaneously for step inputs.

(d) (a) $C_1(2) = 1 - e^{-1}$; (b) $C(2) = 1 - e^{-1} - (0) = 1 - e^{-1}$

6-4. (a) From Problem 4-21: $G(z) = \frac{1.037}{z - 0.7408}$

$$T(z) = \frac{G(z)}{1+G(z)H} = \frac{1.037}{z - 0.7408 + 1.037(0.04)} = \frac{1.037}{z - 0.700}$$

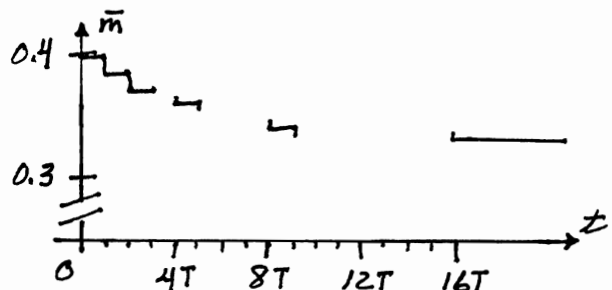
$$C(z) = R(z)T(z) = \frac{0.4z}{z-1} \frac{1.037}{z-0.7} = z \left[\frac{1.383}{z-1} + \frac{-1.383}{z-0.7} \right]$$

$$\therefore C(kT) = \underline{1.383[1 - (0.7)^k]}$$

$\bar{u}(t) = 0.4 \text{ V}$, since the sensor output = 0.4 with $C(t) = 10^\circ \text{C}$.

(b) $m(kT) = 0.4[1 - (0.1)C(kT)]$

k	$C(kT)$	$m(kT)$
0	0	0.4
1	0.416	0.384
2	0.704	0.372
4	1.05	0.360
8	1.30	0.348
16	1.38	0.345
∞	1.383	0.345



(c) $C_{ss}(kT) = \lim_{z \rightarrow 1} (z-1)C(z) = \frac{0.415}{z-0.7} \Big|_{z=1} = \underline{1.383}$

$$6-4.(d) \quad T(z) = \frac{G(z)}{1+G(z)H} ; \quad G(z) = \frac{1.037(\frac{K}{z})}{z-0.7408}$$

$$T(z) = \frac{0.519K}{z-0.7408+0.519K(0.04)} = \frac{0.519K}{z-0.7408+0.0207K}$$

$$\text{dc gain} = T(1) = \frac{0.519K}{0.259+0.0207K}$$

$$\lim_{K \rightarrow \infty} T(1) = \frac{0.519K}{0.0207K} = 25$$

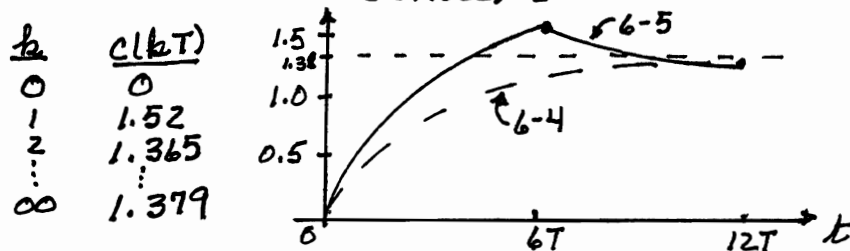
$$\therefore C_{ss}(kT) = (0.4)(25) = 10, \therefore \text{no steady-state error.}$$

$$6-5.(a) \quad G(z) = \left(\frac{z-1}{z}\right) \mathcal{Z}\left[\frac{1}{s(s+0.5)}\right] = \frac{z-1}{z} (2) \frac{2z(1-e^{-3})}{(z-1)(z-e^{-3})} = \frac{3.801}{z-0.04979}$$

$$\therefore T(z) = \frac{G(z)}{1+G(z)H} = \frac{3.801}{z-0.04979+3.801(0.04)} = \frac{3.801}{z+0.1022}$$

$$C(z) = \frac{0.4z}{z-1} \frac{3.801}{z+0.1022} = \frac{1.379z}{z-1} + \frac{-1.379z}{z+0.1022}$$

$$\therefore C(kT) = 1.379[1 - (-0.1022)^k]$$



(b) At $t=0$, $m(t)=1$, and doesn't change until $t=6$.

$$\therefore C(s) = \frac{0.4}{s} \times \frac{z}{s+0.5} = \frac{0.8}{s(s+0.5)} = \frac{1.6}{s} - \frac{1.6}{s+0.5}$$

$$\therefore C(t) = 1.6(1 - e^{-0.5t}), \quad 0 \leq t \leq 6$$

$$6-6.(a) \quad \text{From Problem 6-4, } T(z) = \frac{1.037}{z-0.700}$$

$$\therefore \gamma = \frac{-T}{\ln r} = \frac{-0.6}{\ln(0.7)} = 1.685$$

$$(b) \quad e^{-t/\gamma} = 0.02 \Rightarrow t = -\gamma \ln(0.02) = -(1.68) \ln(0.02) = 6.575 \approx 4T$$

$$(c) \quad \text{From Problem 6-5, (a) } T(z) = \frac{1.90}{z+0.1022}$$

$$\therefore \gamma = \frac{-T}{\ln r} = \frac{-6}{\ln(0.1022)} = 2.635$$

6-6.(c) (b) $t = -T \ln(0.02) = \underline{10.3s} \approx 4T$

(d) $T(s) = \frac{G_p(s)}{1 + G_p(s)H} = \frac{2}{s + 0.5 + 2(0.04)} = \frac{2}{s + 0.58}$

$\therefore T = \frac{1}{0.58} = \underline{1.72s} \Rightarrow e^{-4T} = 0.02, \therefore t = -(1.72) \ln(0.02) = \underline{6.73s} \approx 4T$

6-7.(a) $r(t) = 1.4V$, since the sensor output $= 1.4$ with $c(t) = 20^\circ$

$$G(z) = \left(\frac{z-1}{z}\right)^2 \left[\frac{4}{s^2(s+2)} \right] = \frac{4}{z} \left(\frac{z-1}{z}\right) \left[\frac{z(0.2 - 1 + e^{-0.2}) + (1 - e^{-0.2} - 0.2e^{-0.2})}{2(z-1)^2(z - e^{-0.2})} \right]$$

$$= \frac{0.01873z + 0.01752}{(z-1)(z - 0.8187)}$$

$$T(z) = \frac{KG(z)}{1 + KG(z)H} = \frac{10(0.01873z + 0.01752)}{z^2 - 1.8187z + 0.8187 + 0.01311z + 0.01226}$$

$$= \frac{0.1873z + 0.1752}{z^2 - 1.8056z + 0.8310}$$

$$C(z) = \left(\frac{1.4z}{z-1}\right) T(z) = \frac{1.4z(0.1873z + 0.1752)}{(z-1)(z^2 - 1.8056z + 0.8310)}$$

(b) $C_{ss}(kT) = \lim_{z \rightarrow 1} (z-1)C(z) = \frac{1.4(0.1873 + 0.1752)}{1 - 1.8056 + 0.8310} = \underline{20}$

(c) poles: $z = 0.9028 \pm j0.1262 = 0.9116 \angle \underline{7.96^\circ}$

$\therefore T = -\frac{T}{\ln r} = \frac{-0.1}{\ln(0.9116)} = \underline{1.08s}; t \approx 4T = \underline{4.32s}$

6-8.(a) The open-loop system has the characteristic of $G_p(s)$.

$\therefore$ poles at: $s_1 = 0, \tau_1 = \infty$
 $s_2 = -2, \tau_2 = \underline{0.5s}$

(b) From Problem 6-7, char. eq.: $z^2 - 1.8056z + 0.8310 = 0$

zeros: $z = 0.9116 \angle \underline{7.96^\circ} = 0.9116 \angle \underline{0.139 \text{ rad}}$

$$T = \frac{-T}{\ln r} = \frac{-0.1}{\ln(0.9116)} = \underline{1.08s}$$

$$\omega_n = \frac{1}{T} \sqrt{\ln^2 r + \theta^2} = \frac{1}{0.1} [\ln^2(0.9116) + (0.139)^2]^{\frac{1}{2}} = \underline{1.670}$$

$$\zeta = \frac{-\ln r}{\sqrt{\ln^2 r + \theta^2}} = \underline{0.554}$$

6-8. (c) (a) same as (a)

$$(b) T(s) = \frac{KG_p(s)}{1+KG_p(s)H} = \frac{(10) \frac{4}{s^2+2s}}{1+0.07 \frac{40}{s^2+2s}}$$

$$= \frac{40}{s^2+2s+2.8} = \frac{40}{s^2+2\zeta\omega_n s + \omega_n^2}$$

$$\text{poles: } s = \frac{-2 \pm \sqrt{4-11.2}}{2} = -1 \pm j 1.3416 = 1.673 \angle \pm 126.7^\circ$$

$$\therefore \tau = \frac{1}{\zeta} \quad \zeta = \frac{2}{2(1.67)} = 0.5988$$

$$\omega_n = (2.8)^{1/2} = 1.67$$

$$(d) (b) \zeta = 0.554, -\zeta\pi/\sqrt{1-\zeta^2} = -(0.554)\pi/0.8325 = -2.091$$

$$\therefore \% \text{ overshoot} = (e^{-2.091})100 = 12.4\%$$

$$(c) -\zeta\pi/\sqrt{1-\zeta^2} = -2.349, \% \text{ overshoot} = (e^{-2.349})100 = 9.5\%$$

$$6-9. (a) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{10}{s^3} \right] = 10 \left(\frac{z-1}{z} \right) \frac{(0.1)^2 z(z+1)}{2(z-1)^2} = \frac{0.05(z+1)}{(z-1)^2}$$

(s+a) has the time constant $\tau = 1/a$.

$\therefore \tau = \infty$ at both poles.

The characteristics of the plant are not changed by sampling.

$$(b) T(z) = \frac{KG}{1+KG(z)H_b} = \frac{5(z+1)}{z^2-2z+1+0.1(z+1)} = \frac{5(z+1)}{z^2-1.9z+1.1}$$

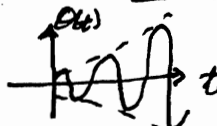
$$\text{poles: } z = 0.95 \pm j 0.4444 = 1.049 \angle \pm 25.07^\circ$$

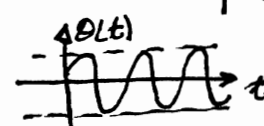
$\therefore$ unstable

$$(c) (a) G_p(s) = \frac{10}{s^2} \quad \therefore \tau = \infty \text{ at both poles}$$

$$(b) T(s) = \frac{KG_p(s)}{1+KG_p(s)H_b} = \frac{1000}{s^2+20}, \text{ poles: } s = \pm j 4.472$$

$\therefore$ marginally stable - $\tau = \infty$ for both poles.

(d) sampled-data: 

analog: 

6-10. (a) From Example 6.2: $G(z) = \frac{(T-1+E^{-T})z + (1-E^{-T}-TE^{-T})}{(z-1)(z-E^{-T})}$

$T=0.5, G(z) = \frac{0.1065z + 0.09020}{(z-1)(z-0.6065)}$

$\therefore$ Char. eq.: $z^2 - 1.6065z + 0.6065 + 0.1065z + 0.09020$
 $= z^2 - 1.500z + 0.6967 = 0$

zeros: $z = 0.75 \pm j0.3663 = 0.8347 / \pm 26.03^\circ = 0.8347 / \pm 0.454 \text{ rad}$

$\gamma = -\frac{T}{\ln r} = 2.7675$

$\omega_n = \frac{1}{T} [\ln^2 r + \theta^2] = 0.9773$

$\beta = \frac{-\ln r}{[\ln^2 r + \theta^2]^{1/2}} = 0.3698$

(b) $T=0.1, G(z) = \frac{0.004837z + 0.004679}{(z-1)(z-0.9048)}$

$\therefore$ Char. eq.: $z^2 - 1.9048z + 0.9048 + 0.004837z + 0.004679$
 $= z^2 - 1.900z + 0.9095$

zeros: $z = 0.950 \pm j0.0837 = 0.9537 / \pm 5.033^\circ$
 $= 0.9537 / \pm 0.0878 \text{ rad}$

$\gamma = \frac{-T}{\ln r} = 2.1095$

$\omega_n = \frac{1}{T} [\ln^2 r + \theta^2] = 0.9978$

$\beta = \frac{-\ln r}{[\ln^2 r + \theta^2]^{1/2}} = 0.4751$

(c) $T(s) = \frac{G_p(s)}{1+G_p(s)} = \frac{1}{s^2+s+1} = \frac{1}{s^2+2\beta\omega_n s + \omega_n^2}$
 $= \frac{1}{(s+0.5)^2 + 0.75}$

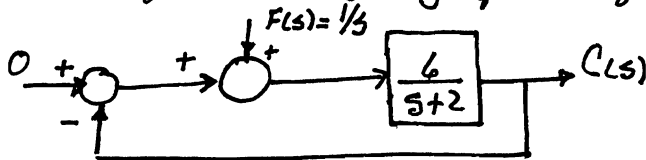
$\therefore \gamma = \frac{1}{0.5} = 2$; $\omega_n^2 = 1 \Rightarrow \omega_n = 1$

$2\beta\omega_n = 1 \Rightarrow \beta = \frac{1}{2(1)} = 0.5$

(d)	$\frac{T}{1}$	γ	ω_n	β	$f_s = 1/T$
	1	4.365	0.9191	0.25	1
	0.5	2.77	0.977	0.369	2
	0.1	2.11	0.998	0.475	10
analog	2		1	0.5	$\rightarrow \infty$

6-10.(d) As the sampling frequency decreases: T increases
 ω_n increases
 ζ decreases

6-11.(a) Can ignore sampling if all signals are constant.



$$C(s) = \frac{G_p(s)}{1 + G_p(s)} F(s) = \frac{6}{s+2} \cdot \frac{1}{s} = \frac{4/8}{s} + \frac{-4/8}{s+2}$$

$$\therefore C_{ss}(t) = \frac{6}{8} = \underline{0.75}$$

$$(b) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{6}{s(s+2)} \right] = \frac{z-1}{z} \left[\frac{6z(1-e^{-0.2})}{(2)(z-1)(z-e^{-0.2})} \right] = \frac{0.5438}{z-0.8187}$$

$$\mathcal{Z} \left[\frac{6}{s(s+2)} \right] = \frac{0.5438z}{(z-1)(z-0.8187)} = \overline{FG}_p(z)$$

$$C(z) = \frac{\overline{FG}_p(z)}{1 + D(z)G(z)} = \frac{\frac{0.5438z}{(z-1)(z-0.8187)}}{1 + \left(\frac{1.12-1}{z-1} \right) \cdot \left(\frac{0.5438}{z-0.8187} \right)}$$

$$= \frac{0.5438z}{z^2 - 1.221z + 0.2749}$$

poles: $z = 0.924, 0.298$ $\therefore$ no poles at $z=1$.

$$\therefore C_{ss}(t) = \lim_{z \rightarrow 1} (z-1)C(z) = \underline{0}$$

$$6-12.(a) (i) T = \frac{-T}{\ln r} = \frac{-0.1}{\ln(0.999)} = \underline{99.95s}, \text{ exponential}$$

$$(ii) T = \frac{-0.1}{\ln(0.99)} = \underline{9.95s}, \text{ exponential}$$

$$(iii) T = \frac{-0.1}{\ln(0.9)} = \underline{0.95s}, \text{ exponential}$$

$$(iv) T = \frac{-0.1}{\ln(0.9)} = \underline{0.95s}; \omega = \frac{\omega_s}{2} = \frac{\pi}{0.1} = 31.42 \text{ rad/s}$$

damped sinusoid

6-12. (a) (v) zeros: $z = 0.965, 0.885$

exponential, $\tau_1 = \frac{0.1}{\ln(0.965)} = 2.81s$; $\tau_2 = \frac{0.1}{\ln(0.885)} = 0.819s$

(vi) zeros: $z=1, -1$ $z=1$ - constant output
 $z=-1$ - sinusoidal output, $\omega = \omega_s/2 = 31.42 \text{ rad/s}$

(vii) zeros: $z=1.01, 0.99$ $\therefore$ unstable

(viii) zeros: $z = 0.837 \angle \pm 44.2^\circ, 0.837 \angle \pm 0.771 \text{ rad}$

$\therefore$ damped sinusoid

$$\tau = \frac{-T}{\ln r} = 0.56s$$

$$\zeta = \frac{-\ln r}{[\ln^2 r + \theta^2]^{1/2}} = 0.156$$

$$\omega_n = \frac{1}{T} [\ln^2 r + \theta^2]^{1/2} = 1.38 \text{ rad/s}$$

(b) (i) $\tau = 999.5s$

(ii) $\tau = 99.5s$

(iii) $\tau = 9.5s$

(iv) $\tau = 9.5s, \omega = 3.142 \text{ rad/s}$

(v) $\tau_1 = 28.1s, \tau_2 = 8.19s$

(vi) $\omega = 3.142 \text{ rad/s}$

(vii) unstable

(viii) $\tau = 5.6s, \omega_n = 1.38 \text{ rad/s}, \zeta = 0.156$

(c) τ, ω_n vary with T , ζ independent of T .

6-13. (a) From Problem 6-1, $T(z) = \frac{0.6321}{z + 0.2642}$

$$\therefore \tau = \frac{-T}{\ln r} = \frac{-1}{\ln(0.2642)} = 0.751s$$

$$(b) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{0.5e^{-0.2Ts}}{s(s+0.5)} \right] = \frac{z-1}{z} \mathcal{Z}_m \left[\frac{0.5}{s(s+0.5)} \right]_{m=0.8}$$

$$= \frac{z-1}{z} \left[\frac{1}{z-1} - \frac{e^{-0.8}}{z-e^{-1}} \right] = \frac{0.5507z + 0.0814}{z(z-0.3679)}$$

$$\therefore \text{char. eq. : } 1 + G(z) = 0 = z^2 - 0.3679z + 0.5507z + 0.0814$$

$$= z^2 + 0.1828z + 0.0814 = 0$$

zeros: $z = 0.2853 \angle \pm 108.7^\circ$

$$\therefore \tau = \frac{-T}{\ln r} = \frac{-1}{\ln(0.2853)} = 0.7973s$$

(c) From Problem 6-1, $G(z) = \frac{0.6321}{z(z-0.3679)}$

$$\text{char. eq. : } z^2 - 0.3679z + 0.6321 = 0$$

6-13.(c) zeros: $z = 0.7950 \angle \pm 76.6^\circ$

$$\tau = \frac{-T}{\ln R} = \underline{4.36s}$$

(d) Delay slows the system response.

6-14.(a) (i) type 0 (ii) type 0 (iii) type 1
(iv) type 2 (v) type 0

(b) (i), (ii), (v) one pole at $z=1$, with the system stable.
(iii), (iv) system stable only

(c) (i), (ii), (v) 2 poles at $z=1$, with the system stable.
(iii) 1 pole at $z=1$, with the system stable.
(iv) system stable only.

6-15.(a) $(1-z^{-1})M(z) = (1-0.9z^{-1})E(z)$

$$D(z) = \frac{M(z)}{E(z)} = \frac{z-0.9}{z-1} \quad \therefore \text{system is type } \underline{1}$$

(b) Type 1 system, $\therefore$ no steady-state error, $\therefore C_{ss}(kT) = \underline{1}$

$$(c) G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s+1}\right] = \frac{z-1}{z} \frac{z(1-e^{-1})}{(z-1)(z-e^{-1})} = \frac{0.6321}{z-0.3679}$$

$$\begin{aligned} \frac{D(z)G(z)}{1+D(z)G(z)} &= \frac{0.6321(z-0.9)}{z^2-1.368z+0.6379+0.6321z-0.5689} \\ &= \frac{0.6321(z-0.9)}{z^2-0.7358z-0.201} \end{aligned}$$

$\therefore$ poles: $z = 0.9479, -0.2121$

$$\tau_1 = \frac{-T}{\ln R} = \frac{-1}{\ln(0.9479)} = \underline{18.7s}, \quad \tau_2 = \underline{0.645s}$$

$\therefore 4\tau_1 = \underline{74.8s}$

$$(d) \frac{C(z)}{z} = \frac{0.6321(z-0.9)}{(z-1)(z-0.9479)(z+0.2121)} = \frac{1}{z-1} + \frac{-0.500}{z-0.9479} + \frac{-0.500}{z+0.2121}$$

$\therefore C(kT) = \underline{1 - (0.9479)^k - 0.5(-0.2121)^k}$

(b) $C_{ss}(kT) = \underline{1}$ (c) $C(75) = 1 - 0.5(0.9479)^{75} - 0.5(-0.2121)^{75}$
 $= 1 - 0.0090 - () \times 10^{-51} \approx \underline{1}$

6-16. (a) From Problem 6-15, type 1

$$(b) K_r = \lim_{z \rightarrow 1} \frac{1}{T} (z-1) D(z) G(z) = \frac{0.6321(1-0.9)}{(1-0.3679)} = \underline{0.1}$$

$$\therefore e_{ss} = \frac{1}{K_r} = \underline{10}$$

$$\therefore e_{ss}(kT) = r_{ss}(kT) - c_{ss}(kT)$$

$$\therefore c_{ss}(kT) = r_{ss}(kT) - e_{ss}(kT) = \underline{kT - 10}$$

(c) From Problem 6-15, time to reach steady-state $\approx \underline{74.85}$

$$(d) \mathcal{Z}[t] = \frac{Tz}{(z-1)^2} = \frac{z}{(z-1)^2}$$

From Problem 6-15:

$$\begin{aligned} \frac{C(z)}{z} &= \frac{0.6321(z-0.9)}{(z-1)^2(z-0.9499)(z+0.2121)} \\ &= \frac{1}{(z-1)^2} + \frac{-10.1}{z-1} + \frac{9.60}{z-0.9499} + \frac{0.413}{z+0.2121} \end{aligned}$$

$$\therefore c_{ss}(kT) = kT - 10.1 \approx kT - 10 \quad (\text{numerical error})$$

$$e_{ss}(kT) = kT - c_{ss}(kT) \approx \underline{10}$$

$$6-17. (a) G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s+1}\right] = \frac{z-1}{z} \frac{z}{z-1} = \frac{z-1}{z-0.3679}$$

$$\therefore D(z)G(z) = \frac{z-0.9}{z-0.3679} \quad \therefore \text{system type} = 0$$

$$(b) D(1)G(1) = \frac{1-0.9}{1-0.3679} = 0.158$$

$$T(1) = \frac{0.158}{1+0.158} = \underline{0.1364} = c_{ss}(kT)$$

$$(c) 1+G(z) = z-0.3679+z-0.9 = 2(z-0.634) = 0$$

$$\therefore T = \frac{-T}{\ln 2} = \frac{-1}{\ln(0.634)} = \underline{2.195}, \therefore 4T = \underline{8.725}$$

$$(d) \frac{D(z)G(z)}{1+D(z)G(z)} = \frac{z-0.9}{z-0.3679+z-0.9} = \frac{z-0.9}{2z-1.2679} = \frac{0.5z-0.45}{z-0.634}$$

$$\therefore \frac{C(z)}{z} = \frac{0.5z-0.45}{(z-1)(z-0.634)} = \frac{0.1366}{z-1} + \frac{0.3634}{z-0.634}$$

$$\therefore c(kT) = 0.1366 + 0.3634(0.634)^k$$

$$\therefore c_{ss}(kT) = \underline{0.1364}$$

$$6-18.(a) \frac{G_p(s)}{1+G_p(s)} = \frac{1}{s+1+1} = \frac{1}{s+2} = \frac{C(s)}{R(s)}$$

$$(s+2)C(s) = R(s)$$

$$\dot{c}(t) + 2c(t) = r(t), \quad r(t) = 1$$

$$\dot{c}(t) = 1 - 2c(t), \quad t \geq 0$$

$$(b) H = 0.25s, \quad c(0) = 0$$

$$c(kH) = c[(k-1)H] + H \{1 - 2 * c[(k-1)H]\}$$

$$= c[(k-1)H] (1-2H) + H = 0.5c[(k-1)H] + 0.25$$

kH	$c(kH)$	$c(kH)$ [exact, (d)]
0	0	0
0.25	0.25	0.197
0.50	0.375	0.316
0.75	0.4375	0.388
1.00	0.4688	0.432
1.25	0.4844	0.459
1.50	0.4922	0.475

$$(c) \frac{C(s)}{\bar{M}(s)} = \frac{1}{s+1} \Rightarrow \dot{c} + c = \bar{m}; \quad \dot{c}(t) = \bar{m}(t) - c(t)$$

$$\bar{m} = 1 - c$$

$$c(kH) = c[(k-1)H] + H \{ \bar{m}[(k-1)H] - c[(k-1)H] \}$$

$$= [1-H] c[(k-1)H] + H \bar{m}[(k-1)H] = 0.75 c[(k-1)H] + 0.25 \bar{m}[(k-1)H]$$

kH	$c(kH)$	$\bar{m}(kH)$	$c(kH)$ (exact (d))
0	0	1	0
0.25	0.25	1	
0.50	0.4375	0.5625	0.3935
0.75	0.4688	0.5312	
1.00	0.4922	0.5078	0.477
1.25	0.4961	0.5039	
1.50	0.4990	0.5010	0.4952

$$(d) (b) C(s) = \frac{1}{s(s+2)} = \frac{1/2}{s} - \frac{1/2}{s+2} \Rightarrow c(t) = \frac{1}{2}(1 - e^{-2t})$$

$$(c) G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{1}{s(s+1)} \right] = \frac{0.3935}{z-0.6065}$$

$$\frac{G(z)}{1+G(z)} = \frac{0.3935}{z-0.2130}$$

$$6-18. (d) \therefore C(z) = \frac{0.3935z}{(z-1)(z-0.2130)} = \frac{0.50z}{z-1} + \frac{-0.50z}{z-0.2130}$$

$$\therefore c(kT) = 0.5 [1 - (0.2130)^k]$$

$$6-19. (a) zX(z) - zX(0) = X(z) - HX(z)$$

$$\therefore [z - (1-H)]X(z) = zX(0)$$

$$X(z) = \frac{zX(0)}{z - (1-H)} \Rightarrow x(kH) = X(0)(1-H)^k$$

$$(b) x(1) = (1)(0.9)^{10} = \underline{0.3487}$$

$$(c) x(1) = (1)(0.99)^{100} = \underline{0.3660}$$

$$(d) (b) \% \text{ error} = \left(\frac{0.3678 - 0.3487}{0.3678} \right) 100 = 5.19\%$$

$$(c) \% \text{ error} = \left(\frac{0.3678 - 0.3660}{0.3678} \right) 100 = 0.489\%$$

$$6-20. (a) \dot{x}(t) = -x(t) + r(t)$$

$$x(kH) = x[(k-1)H] + H\dot{x}[(k-1)H] = x[(k-1)H] + H[-x[(k-1)H] + r[(k-1)H]]$$

$$\therefore x(kH) = (1-H)x[(k-1)H] + Hr[(k-1)H]$$

$$(b) X(z) = (1-H)z^{-1}X(z) + Hz^{-1}R(z)$$

$$\therefore X(z) = \frac{H}{z - (1-H)} R(z) = \frac{H}{z - (1-H)} \cdot \frac{z}{z-1}$$

$$\therefore \frac{X(z)}{z} = \frac{H}{(z-1)[z - (1-H)]} = \frac{1}{z-1} + \frac{-1}{z - (1-H)}$$

$$\therefore x(kH) = \underline{1 - (1-H)^k}$$

$$\therefore x(1) = 1 - (0.9)^{10} = \underline{0.6513}$$

$$(c) x(1) = 1 - (0.99)^{100} = \underline{0.6340}$$

$$(d) 5X(s) + X(s) = \frac{1}{s}$$

$$X(s) = \frac{1}{s(s+1)} = \frac{1}{s} + \frac{-1}{s+1} \Rightarrow x(t) = 1 - e^{-t}$$

$$\therefore x(1) = \underline{0.6321}$$

$$(e) (b) \% \text{ error} = \left| \frac{0.6321 - 0.6513}{0.6321} \right| (100) = \underline{3.037\%}$$

$$(c) \% \text{ error} = \left| \frac{0.6321 - 0.6340}{0.6321} \right| (100) = \underline{0.30\%}$$

$$6-21. \dot{x}(t) = -x(t) \Rightarrow A = -1, B = 0$$

$$(a) (6-32) \quad \dot{x}(k-1) = -x(k-1)$$

$$(6-33) \quad x(k) = x(k-1) - Hx(k-1) = (1-H)x(k-1)$$

$$(6-34) \quad \dot{x}(k) = -x(k) = -(1-H)x(k-1)$$

$$(6-35) \quad x(k) = x(k-1) + \frac{H}{2} [-x(k-1) - (1-H)x(k-1)] \\ = [1 - \frac{H}{2} - \frac{H}{2} - \frac{H^2}{2}]x(k-1) = [1 - H + \frac{H^2}{2}]x(k-1)$$

$$\therefore x(k+1) = (1 - H + \frac{H^2}{2})x(k) = a x(k)$$

$$(b) \quad zX(z) - zX(0) = aX(z) \Rightarrow X(z) = \frac{zX(0)}{z-a}$$

$$\therefore x(kH) = a^k x(0) = (1 - H + \frac{H^2}{2})^k x(0)$$

$$\therefore x(1) = (0.905)^{10} (1) = 0.3685$$

$$(c) \quad 1 - 0.33333 + \frac{(0.33333)^2}{2} = 0.7222$$

$$x(1) = (0.7222)^3 (1) = 0.3767$$

$$(d) \quad sX(s) - x(0) + X(s) = 0$$

$$X(s) = \frac{x(0)}{s+1} \Rightarrow x(t) = x(0)e^{-t}$$

$$\therefore x(1) = 0.3679$$

$$(e) \quad (b) \quad \% \text{ error} = \left| \frac{0.3679 - 0.3685}{0.3679} \right| (100) = 0.163\%$$

$$(c) \quad \% \text{ error} = \left| \frac{0.3679 - 0.3767}{0.3679} \right| (100) = 2.39\%$$

$$6-22. \dot{x}(t) = -x(t) + r(t), \therefore A = -1, B = 1$$

$$(a) (6-32) \quad \dot{x}(k-1) = -x(k-1) + r(k-1)$$

$$(6-33) \quad x(k) = x(k-1) + H[-x(k-1) + r(k-1)] \\ = (1-H)x(k-1) + Hr(k-1)$$

$$(6-34) \quad \dot{x}(k) = (-1)x(k) + r(k) \\ = -(1-H)x(k-1) - Hr(k-1) + r(k)$$

$$(6-35) \quad x(k) = x(k-1) + \frac{H}{2} [-x(k-1) + r(k-1) - (1-H)x(k-1) - Hr(k-1) + r(k)]$$

$$\therefore x(k) = [1 - \frac{H}{2} - \frac{H}{2} + \frac{H^2}{2}]x(k-1) + (\frac{H}{2} - \frac{H^2}{2})r(k-1) + \frac{H}{2}r(k) \\ = a x(k-1) + c r(k-1) + b r(k)$$

$$6-22. (b) (1 - az^{-1})X(z) = (b + cz^{-1})R(z)$$

$$\therefore X(z) = \frac{bz + c}{z - a} R(z)$$

$$a = 1 - 0.1 + \frac{(0.1)^2}{2} = \underline{0.905} ; b = \frac{0.1}{2} = \underline{0.05}, c = \frac{0.1}{2} - \frac{(0.1)^2}{2} = \underline{0.045}$$

$$\therefore \frac{X(z)}{z} = \frac{0.05z + 0.045}{z - 0.905} \cdot \frac{1}{z - 1} = \frac{1}{z - 1} + \frac{-0.95}{z - 0.905}$$

$$\therefore x(k) = 1 - 0.95(0.905)^k$$

$$x(1) = 1 - 0.95(0.905)^{10} = \underline{0.650}$$

$$(c) a = \underline{0.7222}, b = \underline{0.16667}, c = \underline{0.1111}$$

$$\therefore \frac{X(z)}{z} = \frac{0.16667z + 0.1111}{(z - 1)(z - 0.7222)} = \frac{1}{z - 1} + \frac{-0.8332}{z - 0.7222}$$

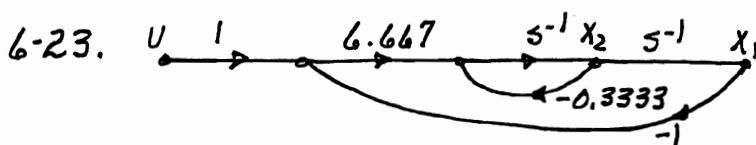
$$\therefore x(k) = 1 - 0.8332(0.7222)^k$$

$$x(1) = 1 - (0.8332)(0.7222)^3 = \underline{0.6862}$$

$$(d) \text{ See Problem 6-20(d), } x(1) = \underline{0.6321}$$

$$(e) \% \text{ error} = \left| \frac{0.6321 - 0.650}{0.6321} \right| (100) = \underline{2.83 \%}$$

$$(c) \% \text{ error} = \left| \frac{0.6321 - 0.6862}{0.6321} \right| (100) = \underline{8.56 \%}$$



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -6.667 & -0.3333 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 6.67 \end{bmatrix} u(t)$$

$$x(k) = x(k-1) + H \dot{x}(k-1)$$

$$\dot{x}(k) = A x(k-1) + B u(k-1)$$

6-23. Computer program - MATLAB

```

A = [0 1; -6.667 -0.3333]; B = [0; 6.667];
u = 1;
H = 0.25;
x = [0; 0];
for k = 0:4;
    t = k*H;
    [t, x(1)]
    xd = A*x + B*u;
    x = x + H*xd;
end

```

results :

<u>Time</u>	<u>output</u>
0	0
0.25	0
0.5	0.4167
0.75	1.2153
1.00	2.1905

*H has been chosen too large.
The system is stable, but
the simulation is
unstable .*